A Hierarchical Gene–Isoform Model for Sparse Single-Cell Data

Purpose

The existing genome-wide factorization models normalized equivalence-class (EC) counts directly. It recovers cell-type information, but a matched evaluation shows a substantial gap from a standard gene-expression analysis. Increasing the factor rank does not close this gap. Adding an auxiliary gene-reconstruction loss helps, indicating that the EC objective spends too much capacity on transcript ambiguity and too little on marker-gene abundance.

This note proposes a hierarchical count model with one cell state controlling both gene expression and splicing. It separates gene abundance from conditional isoform allocation without giving each sparse cell an independent transcript vector or a second latent space. The model is fit jointly across cells and does not require cell-level EM.

Notation

Let $m = 1, \ldots, M$ index cells, $g = 1, \ldots, G$ genes, and $t = 1, \ldots, T$ transcripts. Let $g(t)$ be the gene containing transcript t, and let $\mathcal{T}_g = \{t : g(t) = g\}$. Primer or library type is indexed by p. For primer p, let

$$A_p \in \mathbb{R}_+^{K_p \times T}$$

be the fixed EC-by-transcript observation matrix. It includes alignment compatibility, Salmon conditional weights and positional-bias correction, and the primer-specific transcript sampling exposure. The primary paired model uses no additional transcript-length exposure for oligo(dT) and relative effective-length exposure for random-hexamer priming.

Let n_{kmp} be the observed UMI count for EC k, cell m, and primer p, and let $N_{mp} = \sum_k n_{kmp}$.

One Latent Space, Two Decoder Heads

Each cell has one latent state

$$\boldsymbol{z}_m \in \mathbb{R}^r.$$

Separate decoder loadings map this state to gene abundance and within-gene isoform usage. The same coordinates therefore represent both expression and splicing variation.

Gene abundance

Model the positive abundance of gene g in cell m as

$$\xi_{gm} = \alpha_g + \boldsymbol{u}_g' \boldsymbol{z}_m, \qquad a_{gm} = \exp(\xi_{gm}).$$

Here α_g is a pooled gene intercept and $\boldsymbol{u}_g$ is the gene-expression loading. A softplus link can replace the exponential if it gives more stable gradients.

Conditional isoform usage

For transcript t, define the isoform logit

$$\eta_{tm} = \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m.$$

Normalize only within the transcript's gene:

$$\pi_{tm} = \frac{\exp(\eta_{tm})}{\sum_{j \in \mathcal{T}_{g(t)}} \exp(\eta_{jm})}.$$

The transcript abundance is

$$\theta_{tm} = a_{g(t)m} \pi_{tm}.$$

Thus $\boldsymbol{u}_g$ describes how the shared cell state changes gene abundance, while $\boldsymbol{w}_t$ describes how the same state changes conditional isoform usage. There is no separate splicing state.

The within-gene softmax is an identifiability convention, not a global simplex constraint on Θ. It ensures that a_{gm} is the gene total and that $\boldsymbol{\pi}_{gm}$ contains only within-gene allocation.

Relation to a Flat Multinomial Model

The decomposition

$$\theta_{tm} = a_{g(t)m} \pi_{tm}$$

alone is a reparameterization. Given an unrestricted positive $\boldsymbol{\theta}_m$, one can always recover its gene totals and within-gene proportions. A multinomial likelihood does not make that unrestricted hierarchy more informative than a flat model.

The restriction here is that both parts are generated from the same low-dimensional $\boldsymbol{z}_m$:

$$\log a_{gm} = \alpha_g + \boldsymbol{u}_g' \boldsymbol{z}_m, \qquad \mathrm{logit}_g(\pi_{tm}) = \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m.$$

Cells do not have free a_{gm} or π_{tm}. Sparse cells borrow information through gene and transcript loadings learned from every cell.

This structured model is not generally equivalent to a flat log-linear transcript decoder. Its transcript log abundance is

$$\log \theta_{tm} = \alpha_{g(t)} + \boldsymbol{u}_{g(t)}' \boldsymbol{z}_m + \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m - \mathrm{LSE}_{j \in \mathcal{T}_{g(t)}} (\beta_j + \boldsymbol{w}_j' \boldsymbol{z}_m),$$

whose final gene-specific log-sum-exp term is nonlinear in $\boldsymbol{z}_m$. The restriction disappears only if cell-specific gene totals and isoform logits are allowed enough freedom to reproduce arbitrary $\boldsymbol{\theta}_m$.

Connection to the Paired-Primer Likelihood

For cell m, primer p, and EC k, the unnormalized expected EC mass is

$$q_{kmp} = \sum_t A_{pkt} \theta_{tm} = \sum_t A_{pkt} a_{g(t)m} \frac{\exp(\beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m)}{\sum_{j \in \mathcal{T}_{g(t)}} \exp(\beta_j + \boldsymbol{w}_j' \boldsymbol{z}_m)}.$$

This equation directly connects the shared latent state and both decoder heads to the observed EC counts. Cross-gene ambiguous ECs require no special case: the sum includes every compatible transcript.

Conditioning on the observed UMI total gives

$$\boldsymbol{n}_{mp} \mid N_{mp} \sim \text{Multinomial}\left(N_{mp}, \frac{\boldsymbol{q}_{mp}}{\mathbf{1}'\boldsymbol{q}_{mp}}\right).$$

Ignoring constants, the negative log likelihood is

$$\mathcal{L} = -\sum_{m,p} \sum_{k:n_{kmp}>0} n_{kmp} \log q_{kmp} + \sum_{m,p} N_{mp} \log \zeta_{mp},$$

where

$$\zeta_{mp} = \mathbf{1}'\boldsymbol{q}_{mp}.$$

Define the transcript column-sum exposure

$$d_{pt} = \sum_k A_{pkt}.$$

The normalizer can then be evaluated without predicting every EC:

$$\zeta_{mp} = \sum_g a_{gm} \sum_{t\in\mathcal{T}_g} d_{pt}\pi_{tm}.$$

For a column-normalized oligo(dT) design, $d_{pt} = 1$, so

$$\zeta_{mp} = \sum_g a_{gm}$$

and the normalizer is independent of isoform usage. For the random-hexamer theta design, d_{pt} contains relative transcript exposure, so its normalizer depends on the within-gene isoform mixture and must be evaluated exactly.

The paired primer halves share $\boldsymbol{z}_m$, a_{gm}, and π_{tm}, but use separate A_p. Conditioning on N_{mp} removes a primer-by-cell exposure parameter and preserves the convention of normalizing each primer half independently.

An independent Poisson model,

$$n_{kmp} \sim \text{Poisson}(s_{mp}q_{kmp}),$$

is an alternative when total UMI information should contribute to fitting. It requires estimating or profiling s_{mp}. The conditional multinomial model is the simpler first implementation and matches the sampling unit used for molecule-count folds.

What Must Be Instantiated

The model never stores a complete $T \times M$ matrix Θ. Exact likelihood evaluation does, however, require evaluating transcript logits for the cells in the current minibatch. This is the computational cost of the within-gene softmax; it cannot be projected through A_p as a fixed linear factorization.

For a cell minibatch of size B, the straightforward implementation forms

$$\Xi = \boldsymbol{\alpha} + UZ', \qquad H = \boldsymbol{\beta} + WZ',$$

with shapes $G \times B$ and $T \times B$. A segmented log-sum-exp over gene membership converts H to within-gene Π. The implementation need not form a separate Θ: observed EC predictions gather the required $a_{gm}\pi_{tm}$ values directly, and Z_{mp} uses the segmented weighted sum above. The $T \times B$ logit or probability block is released after the minibatch.

If that workspace becomes limiting, genes can be streamed in blocks. For a fixed cell minibatch:

1. initialize one prediction accumulator for every observed cell–EC event and one normalizer per cell and primer;
2. compute a_{gm} and all transcript logits for a block of complete genes;
3. apply the within-gene softmax;
4. scatter each transcript's contribution through the corresponding sparse rows of A_p into observed-event accumulators;
5. accumulate $a_{gm} \sum_{t \in \mathcal{T}_g} d_{pt}\pi_{tm}$ into ζ_{mp}.

This exact gene-streamed calculation bounds workspace by the largest gene block rather than $T \times B$. A custom autograd operation or a two-pass backward calculation may be needed to avoid retaining every block's graph. The simpler $T \times B$ minibatch implementation should be built first.

Explicit cell-state storage remains small. One million cells with a 32-dimensional float32 state require approximately 128 MB. The dominant cost is decoder evaluation, not storage of cell parameters or Θ.

Why Cell-Level EM Is Not Required

Independent EM would estimate a high-dimensional transcript vector in every cell. That is underdetermined for cells with hundreds or a few thousand UMIs. The hierarchical model instead estimates only r coordinates $\boldsymbol{z}_m$. Gene and transcript intercepts and loadings are shared across all cells.

Cells without molecules that distinguish isoforms remain close to the pooled within-gene baseline $\boldsymbol{\beta}_g$. Informative cells update the shared relationship $\boldsymbol{w}_t'\boldsymbol{z}_m$ between cell state and isoform usage.

EM remains useful on pooled data to initialize β_t. A global or sufficiently large pseudobulk EM estimate has enough molecules to establish baseline isoform usage, but it is not treated as a cell-level estimate. Joint minibatch gradient optimization then estimates $\boldsymbol{z}_m$, gene loadings, and transcript loadings under the complete paired-primer likelihood.

Initialization and Fitting

A staged fit reduces non-convexity and ensures that the latent state first captures robust gene-expression structure:

1. Aggregate ECs that map entirely within one gene and fit $\alpha_g, \boldsymbol{u}_g, \boldsymbol{z}_m$ with a gene-level count factor model. Cross-gene ambiguous ECs are excluded only from this initializer.
2. Estimate pooled transcript proportions from all cells or large pseudobulks and initialize β_t from their within-gene logits.
3. Hold the initialized gene decoder fixed and fit $\boldsymbol{w}_t$ under the complete EC likelihood. The same $\boldsymbol{z}_m$ is used; no new cell variables are introduced.
4. Jointly fine-tune $\boldsymbol{z}_m$, both decoder heads, and both primer observation models under the complete EC likelihood.

Adam or another stable first-order optimizer is appropriate initially, followed by deterministic full-data or large-batch polishing. The gene and isoform decoder heads should have separate learning rates because isoform gradients are supported by fewer molecules. Gradient clipping and segmented log-sum-exp evaluation are required for numerical stability.

The transcript loadings $\boldsymbol{w}_t$ need shrinkage toward zero so genes with little isoform information retain their pooled usage. This is distinct from the removed nuclear-norm penalty on the previous direct factorization: it is a hierarchical prior on deviations from pooled within-gene usage. Its strength must be selected from held-out molecules, not labels.

Identifiability

The following conventions remove avoidable invariances:

- Center the gene intercepts α_g, because a common shift in all gene log abundances is removed by the conditional multinomial likelihood.
- Center β_t within each gene. Adding a common value to every isoform logit in a gene leaves $\boldsymbol{\pi}_{gm}$ unchanged.
- Center every coordinate of $\boldsymbol{w}_t$ within each gene for the same reason.
- Center and standardize columns of Z, absorbing location and scale changes into intercepts and decoder loadings.
- Use an orthogonality convention for reporting. Latent rotations do not affect fitted abundance when both decoder heads rotate with Z, but a stable convention improves warm starts and interpretation.

These are parameter-identification conventions rather than biological regularization or a simplex constraint on the genome-wide transcript matrix.

Label-Blind Model Selection

Molecule counts are partitioned before normalization, as in the existing count-fold procedure. Candidate models are compared using held-out multinomial log likelihood on the same response scale.

The shared rank follows a small-to-large warm-start path, for example

$$r = 4, 8, 16, 32, 64.$$

A one-standard-error rule selects the smallest converged rank statistically indistinguishable from the minimum held-out loss. At each rank, the shrinkage on $\boldsymbol{w}_t$ is selected by an inner strong-to-weak path. The gene-only endpoint $\boldsymbol{w}_t = 0$ is included explicitly.

Held-out diagnostics decompose performance into:

- gene-unambiguous molecule likelihood;
- within-gene EC likelihood;
- cross-gene ambiguous EC likelihood;
- poly(dT) and random-hexamer likelihoods.

This distinguishes improvement in isoform prediction from improvement that comes only through gene abundance. Cell-type labels, ARI, NMI, and silhouette remain external assessments and are never hyperparameter inputs.

Downstream Differential Isoform Analysis

The fitted π_{tm} is the conditional isoform usage needed downstream. Differential effects can enter the same within-gene logits:

$$\eta_{tm} = \beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m + \boldsymbol{x}_m' \boldsymbol{\gamma}_t,$$

where $\boldsymbol{x}_m$ contains diagnosis, cell type, donor covariates, and interactions. Coefficients $\boldsymbol{\gamma}_t$ are centered within each gene.

Inference must respect donor replication, either through donor-level random effects or by fitting and testing donor-by-cell-type pseudobulks of model-implied sufficient statistics. The latent state remains shared: differential splicing is a covariate effect in the isoform decoder, not a second cell representation.

Recommended First Implementation

The first implementation is

$$\boldsymbol{z}_m \in \mathbb{R}^r, \qquad a_{gm} = \exp(\alpha_g + \boldsymbol{u}_g' \boldsymbol{z}_m),$$

$$\pi_{tm} = \mathrm{softmax}_{t \in \mathcal{T}_{g(t)}}(\beta_t + \boldsymbol{w}_t' \boldsymbol{z}_m),$$

$$q_{kmp} = \sum_t A_{pkt} a_{g(t)m} \pi_{tm}.$$

It uses one latent space, a paired-primer multinomial likelihood, pooled-EM initialization of $\boldsymbol{\beta}$, explicit cell states, and GPU cell minibatches. The initial implementation may instantiate a temporary $T \times B$ logit block, but never a $T \times M$ abundance matrix. Gene-streamed likelihood evaluation is an optimization to add only if the minibatch workspace or decoder cost is limiting.

This model tests the central hypothesis directly: one cell state can recover standard gene-expression structure while shared transcript loadings extract the isoform information supported across many sparse cells.